HEATSHIELD — Complete Project Guide

Oct 14, 2026 · @Amisha

The one-minute version

HEATSHIELD is a medication-aware digital twin that predicts whether a vulnerable patient's heat strain will become dangerous in the next 24 hours, and lets a doctor test what-if scenarios before it happens.

- **Who it's for:** older patients with hypertension, type 2 diabetes or reduced kidney function, on medicines that make heat more dangerous (diuretics, SGLT2 inhibitors, beta-blockers).
- **What it predicts:** whether the patient's Physiological Strain Index (PSI, a published 0–10 heat-strain score) will cross 7 ("high strain") in the next 24 hours.
- **What it fuses:** synthetic patient records (EHR), a simulated wearable stream (heart rate, skin temperature, steps, sleep) and real weather from Indian heatwave days.
- **The hook:** standard methods estimate core body temperature from heart rate. Beta-blockers stop heart rate from rising, so those methods underestimate strain in exactly the patients most at risk. HEATSHIELD reads the medication list and adjusts how it interprets the wearable.
- **Why it's a twin, not a classifier:** it learns each patient's own body from their data, forecasts their future under tomorrow's weather, and simulates interventions (stay indoors, AC available, hotter day, medication scenario).
- **What it does not do:** diagnose dehydration or kidney injury, or recommend medication changes. All results are on synthetic data.

Competition: Happiest Health Digital Twin Challenge 2026 (Unstop). Code due 20 Oct 2026, 7 PM IST. Registration closes 18 Oct.

Background: the problem we're solving

What a digital twin actually is

A digital twin is a virtual copy of one specific patient that keeps updating as new data arrives, and can be run forward to answer "what happens next, and what if we change something?"

The difference from a normal ML model:

	Normal ML model	Digital twin
Learns	Patterns across a population	This one patient's body
Output	A risk score right now	A forecast trajectory over time
Updates	Retrained occasionally	Every time new data arrives
Can answer "what if?"	No	Yes, by simulating changed inputs

Most teams in this challenge will build the left column with a dashboard on top. We build the right column.

Why heat is a health problem

The body keeps its core temperature near 37°C. In heat it does two things: it pumps more blood to the skin (so heart rate rises) and it sweats (so it loses water and salt). Heat illness happens when heat comes in faster than the body can get rid of it, or when sweating drains fluid faster than it is replaced.

Older people and people with chronic disease cope worse: they sweat less, feel thirst less, and have less reserve in the heart and kidneys. Indian heatwaves routinely push daytime temperatures into the mid-40s °C, and nights often stay hot, so the body never fully recovers.

Why medications make it worse

This is the core of our idea. Several very common drugs for hypertension and diabetes change how the body handles heat:

Drug class	Common examples	What it does in heat
Diuretics	Hydrochlorothiazide, furosemide	Increase urine output, so the patient starts with less fluid and dehydrates faster
SGLT2 inhibitors	Dapagliflozin, empagliflozin	Lower glucose by passing it into urine, which pulls water with it; adds to fluid loss
Beta-blockers	Metoprolol, atenolol	Blunt the heart-rate rise, which limits heat transfer to the skin and hides the usual warning sign
ACE inhibitors / ARBs	Ramipril, telmisartan	Reduce the kidney's ability to protect itself when fluid is low

In the prototype we model the first three explicitly. ACE inhibitors/ARBs can be listed as a kidney-vulnerability flag only.

The gap we fill

Personal heat-strain twins already exist, but for healthy workers, soldiers and athletes. A systematic review of 25 wearable core-temperature algorithms found few of them used individual health data. Nobody we found builds one for chronically ill patients using their medical record. That is HEATSHIELD's contribution: a heat-strain twin that is personalised by the EHR, especially the medication list.

Key concepts you need to understand

1. Core temperature vs skin temperature

Core temperature (Tc) is the temperature deep inside the body; it is what actually matters for heat illness. Normal is about 37°C; above roughly 39.5°C is dangerous. Skin temperature (Tsk) is what a wrist wearable measures; it swings with the weather and does not equal core temperature.

The problem: consumer wearables cannot measure core temperature. So we must **estimate** it from what they can measure.

2. Physiological Strain Index (PSI)

PSI (Moran et al., 1998) combines the rise in core temperature and the rise in heart rate into one 0–10 score:

```
PSI = 5 \cdot \frac{T_{c,t} - T_{c,0}}{39.5 - T_{c,0}} + 5 \cdot \frac{HR_t - HR_0}{180 - HR_0}
```

Tc,0 and HR0 are the patient's resting baseline (measured at rest, e.g. early morning). Each half contributes up to 5 points. Commonly used bands: 0–2 none/little, 3–4 low, 5–6 moderate, 7–8 high, 9–10 very high strain.

Our prediction target: **will PSI reach 7 or more at any point in the next 24 hours?**

Note: for a beta-blocked patient, the HR half of PSI is suppressed too. So in the twin we also report the temperature half on its own, which is a more honest strain signal for those patients.

3. The Kalman filter, in plain words

A Kalman filter estimates something you cannot see (here: core temperature) from noisy things you can see (here: heart rate). At every time step it does two things:

1. **Predict:** "Based on my model, core temp should now be about X, give or take some uncertainty."
2. **Correct:** "Heart rate just came in. If my model is right, HR should be Y. It's actually Z, so I'll nudge my estimate of core temp toward what explains Z."

The output is a running estimate **plus** an uncertainty band. That uncertainty is exactly what we show the doctor.

4. The Buller method (heart rate to core temperature)

Buller et al. (2013) published a Kalman filter that estimates core temperature from heart rate alone. Its pieces:

- **State model:** core temperature changes slowly, as a random walk: Tc(t) = Tc(t−1) + noise.
- **Observation model:** heart rate is a fixed curved function of core temperature (a quadratic fitted on soldiers and athletes).

It works well for healthy young people exercising. It is our baseline method, and its exact coefficients must be checked against the paper (see the last section).

5. Why beta-blockers break it

The Buller observation model assumes "if core temp is high, heart rate will be high." A beta-blocker breaks that link: core temperature rises, but heart rate stays lower than the curve expects. The filter then concludes core temperature is lower than it really is, and **underestimates strain** in exactly the patients most at risk.

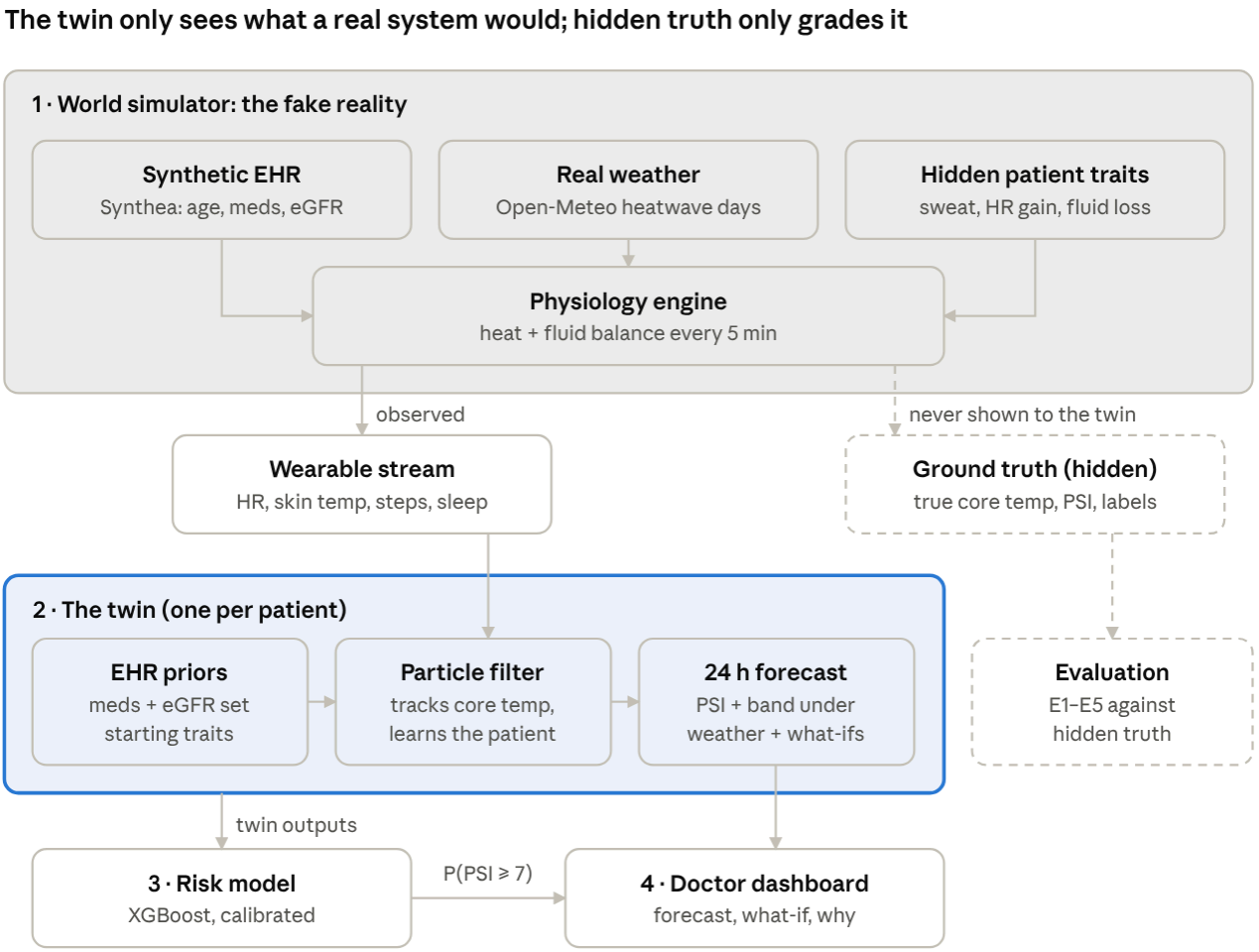
HEATSHIELD fixes this in three ways:

1. It reads the medication list from the EHR and starts with a weaker HR-to-temperature link for beta-blocked patients.
2. It adds skin temperature and the weather as extra evidence, so it doesn't rely on heart rate alone.
3. It replaces the random-walk state model with a simple heat-balance model driven by weather and activity, so it knows when core temperature *should* be rising even if heart rate doesn't show it.

This gives us our headline experiment: **standard method vs medication-aware twin, on beta-blocked vs non-beta-blocked patients.**

Architecture overview

HEATSHIELD has four components: a world simulator that plays reality, the twin that learns each patient, a risk model, and the doctor dashboard.



HEATSHIELD architecture · 4 components

The twin receives only the EHR (as priors), the wearable stream and the weather. The hidden traits and ground truth (dashed) never reach it; they are used only to grade it, which is what avoids the circularity problem.

Component 1: the world simulator (the fake reality)

The world simulator creates synthetic patients and their minute-by-minute physiology during real heatwave days, including a "true" core temperature the twin never gets to see. It plays the role of the real world, so we can check the twin against ground truth.

Why we need a separate simulator: if the same equations generate the data and run the twin, the twin is just reading its own answers back (the circularity problem). So the world simulator is deliberately **richer** than the twin: it has hidden per-patient traits, sensor noise, and effects the twin doesn't model.

1a. Patient records (EHR)

Source: **Synthea**, an open-source synthetic patient generator (Java). Run it, then filter to adults aged 55+ with hypertension and/or type 2 diabetes and/or chronic kidney disease. From its CSV output we keep:

EHR field	Synthea file	Used for
Age, sex	patients.csv	Sweat capacity, baseline HR
Diagnoses (hypertension, T2D, CKD)	conditions.csv	Risk group, priors
Medications	medications.csv	Drug flags: diuretic, SGLT2i, beta-blocker, ACEi/ARB
Creatinine / eGFR, BMI, BP	observations.csv	Kidney reserve, body mass for heat capacity

Two practical caveats to handle and document:

- Synthea's default population is US-based. We use it for realistic record *structure* and disease/medication combinations, then re-sample age and add India-relevant fields ourselves.
- Synthea may not prescribe every drug class (e.g. SGLT2 inhibitors) often enough. We top up medication flags to get a balanced cohort, and say so.

Fallback if Synthea eats more than half a day: write a small Python generator that outputs the same columns. Judges care that the EHR is realistic and documented, not which tool made it.

We add two India-specific fields: **home has AC (yes/no)** and **outdoor work or errands (yes/no)**. These drive exposure and become what-if levers.

Target cohort: about 200 patients, roughly half on a beta-blocker, with overlapping diuretic and SGLT2i use.

1b. Hidden traits (never shown to the twin)

Each patient gets a few personal parameters, drawn from ranges that depend on their age and medications:

Hidden trait	What it means	Shifted by
Sweat capacity	How much heat they can lose by sweating	Lower with age
HR gain	How much heart rate rises per °C of core temp	Much lower on beta-blockers
Fluid-loss rate	How fast they dehydrate in heat	Higher on diuretics and SGLT2i
Fluid reserve	How much loss they tolerate before strain climbs	Lower with low eGFR
Resting HR, resting Tc	Personal baseline	Random per patient

All effect sizes here are **illustrative assumptions**, written down in one config file and in the README. We never present them as clinical facts.

1c. Weather

Source: **Open-Meteo historical weather API** (free, no key). Hourly air temperature, relative humidity and solar radiation for several Indian cities across a recent pre-monsoon heatwave (April–June). Download once and save as CSV in the repo, so the demo never depends on the internet.

Indoor temperature is derived from outdoor: it lags and dampens the outdoor curve, and is capped lower if the home has AC. A "power cut" scenario removes the cap.

1d. Daily routine

Each patient gets a simple schedule: sleep at night, a morning walk for some, indoor rest at midday, outdoor errands for those flagged "outdoor". The schedule sets activity level (in METs) and whether they're exposed to outdoor or indoor temperature at each

moment.

1e. Physiology engine

At 5-minute steps, a simple heat-and-fluid balance updates each patient:

- **Heat in:** metabolic heat from activity, plus heat from the environment when it is hotter than the skin.
- **Heat out:** sweating (limited by sweat capacity, humidity and how dehydrated they are) plus passive loss when the environment is cooler.
- **Core temperature** rises or falls with the balance, scaled by body mass.
- **Fluid deficit** grows with sweating and medication-driven loss, and is replaced by drinking at meal and routine times.
- **Heart rate** = resting HR + an activity term + HR gain × core-temperature rise + a dehydration term. The beta-blocker effect lives in HR gain.
- **Skin temperature** follows a blend of environment and core temperature.

1f. Outputs

For each patient, a time series every 5 minutes containing:

- What a wearable would see (**given to the twin**): heart rate, skin temperature, steps, sleep state, each with realistic noise and occasional dropouts.
- Ground truth (**hidden from the twin, used only for evaluation and labels**): true core temperature, fluid deficit, true PSI.

Simulate 7 days per patient: days 1–5 for calibration, days 6–7 held back for the hide-and-reveal test.

Component 2: the twin (the core of the project)

The twin is one living object per patient that estimates their hidden state, learns their personal traits, and forecasts their next 24 hours under any scenario. It only ever sees what a real system would see: the EHR, the wearable stream and the weather.

2a. What the twin holds (its state)

Each patient's twin is a Python object containing:

- **Profile** from the EHR: age, conditions, eGFR, medication flags, AC, outdoor flag.
- **Estimated hidden state:** current core temperature and fluid deficit, each with an uncertainty.

- **Estimated personal traits:** sweat capacity, HR gain, fluid-loss rate, each with an uncertainty.
- **Recent history:** a rolling window of observations.
- **Last forecast** and the scenario it was run under.
- **Last updated** timestamp, and a staleness flag if data stops arriving.

2b. Step 1: priors from the EHR

Before seeing any wearable data, the twin sets starting guesses for the personal traits using the EHR. For example: on a beta-blocker → start with a low HR gain; on a diuretic or SGLT2i → start with a higher fluid-loss rate; low eGFR → smaller fluid reserve; older → lower sweat capacity.

This is where the EHR genuinely changes how the wearable is read, which is the fusion the brief asks for.

2c. Step 2: tracking the hidden state (the filter)

Every 5 minutes, when new wearable data arrives, the twin runs a filter step:

- **Predict** with its own simplified heat-balance model (weather + activity → expected change in core temperature and fluid).
- **Correct** using what it observes: heart rate and skin temperature, each with its own noise level.

Implementation choice: a **particle filter** (a few hundred particles per patient). Each particle is one guess of "current state + personal traits". Particles that explain the observations well survive; bad ones die off. This handles the non-linear physiology and learns the traits at the same time, without us deriving Jacobians. NumPy only, vectorised.

If the particle filter is too slow or unstable by end of 16 Oct, fall back to an extended Kalman filter for the state plus a simple grid search for the traits on the calibration days.

2d. Step 3: personalisation (learning the patient)

Over days 1–5, the particle cloud for each trait narrows. On the dashboard this shows as "the twin's estimate of this patient's sweat capacity, with its confidence" tightening over time. That narrowing is visible proof the twin is learning this specific patient.

We also keep a **population twin**: the same model with average traits and no learning. Comparing personal vs population is one of our key experiments.

2e. Step 4: forecasting under scenarios

To forecast, the twin takes the current particle cloud and runs every particle forward 24 hours using:

- tomorrow's weather (forecast, or a saved heatwave day),
- the patient's expected routine,
- any scenario changes the doctor selected.

The spread of the particles gives the forecast uncertainty band for free. From each trajectory we compute PSI, so we get both a **PSI curve with a band** and **the probability that PSI reaches 7**.

2f. The scenarios (what-if levers)

Scenario	What it changes in the forecast	Type
Hotter day (+1 to +4°C)	Outdoor temperature	Environment
Power cut / no AC	Removes indoor cap	Environment
Stay indoors 11am–4pm	Exposure schedule	Behaviour
Lighter activity	Activity level	Behaviour
Extra fluids at set times	Fluid replacement	Behaviour
Medication scenario (e.g. "if not on a diuretic")	Fluid-loss prior	Hypothetical, clinician review

The medication scenario is always labelled "hypothetical: for clinical judgement, not a recommendation". It shows how much of this patient's risk is attributable to the medication; the decision stays with the doctor.

2g. Explanation

For every alert, the twin answers "why" by comparing the forecast with copies where one factor is reset to average (weather to a normal day, medications off, traits to population average). The factor whose reset lowers risk most is the main driver. Shown as a short ranked list: "Main drivers: forecast 45°C afternoon; diuretic-related fluid loss; low sweat capacity learned from last 5 days."

Component 3: the risk model

The risk model is a small, fast XGBoost classifier that answers one yes/no question each evening: will this patient's true PSI reach 7 in the next 24 hours? It complements the twin's simulation with a learned, calibrated probability.

3a. Why have both a twin forecast and an ML model?

The twin's forecast is mechanistic: it is only as good as its simplified physics. The ML model learns from many patients what actually led to high strain in the world simulator, including effects the twin's physics doesn't capture. Feeding twin outputs into the model and comparing with/without them shows whether the twin adds information. That is a clean, honest experiment.

3b. Label

For each patient and each evening: **1 if the true PSI (from the world simulator's hidden core temperature and HR) reaches 7 at any point in the next 24 hours, else 0.**

Check the class balance early. If almost nobody reaches 7, use cohort and weather days that create enough events, or report the result at PSI ≥ 6 as well.

3c. Features

Group	Examples
EHR	Age, eGFR, diuretic, SGLT2i, beta-blocker, ACEi/ARB, AC at home, outdoor flag
Recent wearable	Last-24h mean and max HR, HR above personal baseline, skin temp max, steps, sleep hours
Weather	Tomorrow's max temperature, max heat index, hours above 40°C, night-time minimum
Twin outputs	Estimated core temp now, estimated fluid deficit, learned sweat capacity / HR gain / fluid-loss rate, twin's forecast P(PSI ≥ 7)

3d. Models compared

1. **Heat-index rule:** alert if tomorrow's heat index exceeds a fixed threshold. What most heat warnings do today.
2. **XGBoost without twin features:** EHR + wearable + weather only.

3. **Twin forecast alone:** the particle forecast's probability.
4. **XGBoost with twin features:** the full HEATSHIELD model.

3e. Training rules

- **Split by patient**, not by row: train ~70% of patients, validate ~15%, test ~15%. Otherwise the same patient appears in train and test and scores are inflated.
- Keep XGBoost small (shallow trees). Calibrate probabilities on the validation set (isotonic or Platt).
- Fix random seeds and save the trained model file, so results reproduce.

3f. What we report

AUROC, AUPRC (better when events are rare), Brier score, a calibration plot, and sensitivity/precision at the chosen alert threshold. Report everything separately for **beta-blocked vs not** and **with vs without AC**. Report the numbers we actually get, even if modest.

Component 4: the doctor dashboard

The dashboard is a Streamlit app with a patient list and five tabs per patient, designed so a judge can change an input and immediately see the virtual patient evolve differently.

Home: patient list (triage view)

A table of all monitored patients, sorted by tomorrow's risk: name/ID, age, key medications as small tags, tomorrow's max temperature, P(PSI ≥ 7), and a colour band (low / moderate / high). Clicking a row opens that patient.

Tab 1: Patient overview

The EHR at a glance: age, conditions, medication list with heat-relevant drugs highlighted, eGFR trend, AC and outdoor flags. One line in plain English: "Heat-relevant: on hydrochlorothiazide (fluid loss) and metoprolol (blunted heart rate)."

Tab 2: Live twin state

- Last 48 hours: observed heart rate and skin temperature (solid lines), estimated core temperature (dashed, with uncertainty band), weather in the background.
- Learned personal traits as small gauges with confidence ranges, plus how they've narrowed over the calibration days.
- Data freshness indicator ("last reading 4 min ago"; turns amber if stale).

Tab 3: 24-hour forecast

- PSI forecast curve with uncertainty band, the 7 threshold as a line, and the high-risk window shaded (e.g. "1:30–4:00 pm").
- P(PSI ⩾ 7) from both the twin and the risk model, shown side by side.
- "Why" panel: the ranked main drivers from 2g.

Tab 4: What-if simulator (the showpiece)

Controls on the left (the levers from 2f); on the right, the baseline forecast and the scenario forecast drawn on the same chart, with the change in peak PSI and in risk shown as a number ("Stay indoors 11–4: peak PSI 7.8 → 5.9, risk 64% → 21%"). These example numbers are placeholders; the app shows real model outputs. A second mode compares **two patients under the same heatwave**.

Tab 5: Evidence and limits

What data each number came from, labelled **Observed** / **Estimated** / **Predicted** / **Hypothetical**; model versions; key assumptions; and the standing notice: "Prototype on synthetic data. Not validated for clinical use." Making this a feature rather than fine print signals scientific maturity to the jury.

Design rules

- Consistent line styles everywhere: solid = observed, dashed = estimated, shaded = uncertainty, dotted = hypothetical.
- One accent colour for risk; everything else neutral.
- Every chart fully works offline from saved data.

Experiments and evaluation

Five experiments, each answering a question a judge will ask; together they prove HEATSHIELD is a twin and not just a dashboard. Run all of them on held-out test patients only.

#	Question	Experiment	Metric	What "success" looks like
E1	Does medication-awareness matter?	Core-temp estimation: Buller baseline vs HEATSHIELD twin, split by beta-blocker yes/no	RMSE of estimated vs true core temp (°C)	Buller error much larger on beta-blocked patients; twin closes the gap
E2	Does the twin really learn the patient?	Personal twin vs population twin	Forecast error on days 6–7; trait-estimate error vs hidden truth	Personal twin beats population; trait estimates converge
E3	Can it forecast days it never saw?	Hide-and-reveal: calibrate days 1–5, forecast 6–7, overlay truth	Forecast RMSE; % of truth inside the uncertainty band	Truth mostly inside the band (aim roughly 80–95% for a 90% band)
E4	Does it predict danger better than simple rules?	Compare the 4 models from 3d	AUROC, AUPRC, Brier, calibration	Full model beats heat-index rule and no-twin XGBoost
E5	Do scenarios behave sensibly?	Sanity checks on the what-if engine	Pass/fail tests	Hotter day → never lower risk; staying indoors → never higher; AC removed → higher

E1 is the headline result: put it first in the deck and video. E5 should be written as automated tests (pytest), which also shows engineering discipline.

Honest framing for every result: "on synthetic data generated by our world simulator, under stated assumptions". The point is to demonstrate the method and architecture, not clinical accuracy.

Tech stack and repo structure

Everything is plain Python, chosen so you can build fast with tools you already know; no database server and no Docker needed for the submission.

Layer	Tool	Why
Language	Python 3.11	One language end to end
Synthetic EHR	Synthea (Java), with a Python fallback generator	Brief explicitly suggests it
Weather	Open-Meteo historical API, saved to CSV	Free, no key, works offline after download
Data	pandas + Parquet files	Fast, simple, no DB setup
Physiology + filter	NumPy (vectorised particle filter)	Full control, no black box
Risk model	XGBoost + scikit-learn (calibration, metrics)	Strong, small, explainable
Dashboard	Streamlit + Plotly	Working UI in a day
Tests	pytest	Scenario sanity checks (E5)
Diagrams	draw.io or Excalidraw, exported to PDF	Required architecture PDF
License	MIT	Simple, permissive

In the architecture diagram, show PostgreSQL/TimescaleDB, a FHIR API and a streaming broker as the **production path** (dotted boxes), so judges see you know how it would scale without you building it.

Repo layout

Folder name required by the rules: `TeamName_CollegeName`.

```
TeamName_CollegeName/
├── README.md                  # every section the brief requires
├── LICENSE                    # MIT
├── requirements.txt
├── config/
│   └── assumptions.yaml       # every physiological assumption, in one place
├── data/
│   ├── raw/synthea/           # Synthea CSV output (or generator output)
│   ├── raw/weather/           # saved Open-Meteo CSVs
│   └── processed/             # cohort.parquet, wearables.parquet,
truth.parquet
├── src/heatshield/
│   ├── ehr.py                 # load/clean EHR, medication flags, priors
```

```
|   ├── weather.py             # download + load weather, indoor model
|   ├── world/
|   |   ├── physiology.py      # rich heat + fluid engine (ground truth)
|   |   └── simulate.py        # routines, noise, writes wearable + truth
|   ├── twin/
|   |   ├── model.py           # twin's simplified physics
|   |   ├── filter.py          # particle filter + Buller baseline
|   |   ├── twin.py            # PatientTwin class: update, forecast, explain
|   |   └── scenarios.py       # what-if definitions
|   ├── psi.py                 # PSI calculation
|   └── risk/
|       ├── features.py
|       └── train.py           # XGBoost, calibration, metrics
├── experiments/               # scripts for E1–E5, write results/*.csv +
plots
├── results/                   # saved metrics and figures used in the deck
├── app/
|   └── dashboard.py           # Streamlit app
├── tests/
|   └── test_scenarios.py      # E5 sanity checks
└── docs/
    ├── architecture.pdf
    └── presentation.pdf
```

One command to reproduce everything (e.g. `make all` or `python run_all.py`), and one to launch the app (`streamlit run app/dashboard.py`). Judges value a repo that just runs.

Day-by-day plan to 20 Oct

The plan front-loads the risky parts: the world simulator and the twin's filter must work by the evening of 16 Oct, or we simplify. Tick items off as you go.

Wed 14 Oct: setup and data

- [ ] Register on Unstop (deadline 18 Oct, 7 PM) and decide team name
- [ ] Create the GitHub repo with the layout above, MIT license, requirements.txt
- [ ] Verify the PSI formula and Buller coefficients against the papers (last section)
- [ ] Download and save heatwave weather CSVs from Open-Meteo
- [ ] Run Synthea (or start the Python fallback) and build the filtered cohort table

Thu 15 Oct: world simulator

- [ ] Write config/assumptions.yaml with every parameter range

- [ ] Implement the physiology engine and daily routines
- [ ] Generate 7 days × ~200 patients; save wearable and truth Parquet files
- [ ] Plot 3–4 patients and sanity-check: core temp rises on hot afternoons, beta-blocked HR stays flatter, diuretic patients dehydrate faster

Fri 16 Oct: the twin (go/no-go)

- [ ] Implement the Buller baseline filter
- [ ] Implement the twin's simple physics and the particle filter with EHR priors
- [ ] Run E1 (core-temp error, beta-blocked vs not) and E3 (hide-and-reveal)
- [ ] **Gate:** if the particle filter isn't working by tonight, switch to the EKF + grid-search fallback

Sat 17 Oct: forecasting, scenarios, risk model

- [ ] 24-hour forecast with uncertainty, scenario engine, driver explanation
- [ ] E5 sanity tests in pytest
- [ ] Build features, train the 4 models, calibrate; run E2 and E4; save results and figures

Sun 18 Oct: dashboard

- [ ] Patient list + Tabs 1–5, wired to saved data and live twin forecasts
- [ ] Two-patient comparison mode
- [ ] Confirm registration is complete before 7 PM

Mon 19 Oct: polish and documents

- [ ] Fresh clone + one-command run test on a clean environment
- [ ] README with every required section and real results
- [ ] Architecture diagram PDF
- [ ] Presentation PDF (~12–15 slides)
- [ ] Write the video script

Tue 20 Oct: video and submit (deadline 7 PM)

- [ ] Record the 20+ minute video by early afternoon; upload as unlisted YouTube
- [ ] Add the link to the README; make sure the repo is public
- [ ] Submit team leader's name, phone, email and the repo link on Unstop with a few hours to spare

If anything slips, cut in this order: two-patient comparison → explanation panel → AC/power-cut scenario. Never cut E1, the hide-and-reveal, or the video.

Submission checklist and video script

The submission is a public GitHub repo whose README contains everything the brief lists; missing any item can cost the shortlist, so check each one.

README must contain

- [ ] Team details and college/incubator
- [ ] Project title
- [ ] Problem statement and healthcare use case
- [ ] Technical stack, AI/ML model and framework details
- [ ] Link to the video (minimum 20 minutes; unlisted YouTube is allowed)
- [ ] Open-source license details
- [ ] Architecture diagram (PDF/PPT) in the repo, linked
- [ ] Presentation (PDF/PPT) with project details and outcome, linked
- [ ] Repo is public and all links open without login
- [ ] On Unstop: team leader's name, phone, email and the repo link only

Also worth adding: how to run in two commands, a results table with the real numbers, and a clearly marked "Assumptions and limitations" section.

Video script (aim for about 22 minutes)

The brief requires at least 20 minutes, so plan 22 to be safe. Record the screen with your voice; show the working app for most of it.

Time	Segment	What to show
0:00–2:00	The problem	Indian heatwaves; why older patients on common drugs are at higher risk; the beta-blocker blind spot
2:00–4:00	What a digital twin is	Classifier vs twin table; our one-line pitch
4:00–7:00	Data and world simulator	Synthea EHR, Open-Meteo weather, simulated wearables; why a separate world avoids circularity; assumptions file
7:00–10:00	How the twin works	EHR priors, particle filter, learning traits, forecasting; architecture diagram

Time	Segment	What to show
10:00–16:00	Live demo	Patient list → open a high-risk patient → live state → forecast → what-if (stay indoors, hotter day, power cut, medication scenario) → two patients, same heatwave
16:00–19:00	Results	E1 headline, hide-and-reveal overlay, personal vs population, model comparison, scenario tests
19:00–21:00	Limitations and responsible use	Synthetic data only, no clinical claims, no medication advice, DPDP compliance by design
21:00–22:00	Path to real use	Real wearables, FHIR EHR integration, pilot with clinicians, Happiest Health clinics as a natural setting

Do one full dry run before recording. Record in segments if needed and join them; that's fine and less stressful than one take.

Limitations, honest framing, and likely judge questions

Being upfront about limits is a strength with a medical jury; the risky move is overclaiming.

Limitations to state openly

- All data is synthetic. Results show the method works on our simulated world, not that it is clinically accurate.
- The physiology is simplified and the drug effect sizes are illustrative assumptions, documented in one file.
- PSI was designed for exertional heat strain in healthy people; we use it as a research target, not a clinical diagnosis.
- The medication scenario shows attributable risk only. It is not a recommendation to stop or change any drug.
- Real deployment would need clinical validation on real wearable and EHR data, with ethics approval.

Questions you'll likely get, and how to answer

"Isn't your model just learning your own simulator?" No: the world simulator is richer than the twin and has hidden traits the twin never sees. The twin must infer them from observations, and we score it against hidden ground truth on days it never saw.

"Why not a deep learning model?" With synthetic data, a big model would just memorise the simulator. A physics-informed filter is interpretable, gives honest uncertainty, and can run what-if scenarios, which a black box can't.

"How would this work with real patients?" Swap the world simulator for real streams: a smartwatch API for heart rate and skin temperature, a FHIR feed for the EHR, a weather API for forecasts. The twin code doesn't change. A pilot would validate against clinic outcomes such as heat-related visits.

"Why heat? The brief suggested diabetes or hypertension." Our patients *are* hypertensive and diabetic. Heat is the trigger that turns their medications into a risk; that's the adverse event we predict.

"What about privacy (DPDP Act)?" No real data was used. By design the twin only needs data the patient already shares with their doctor, and could run on-device or on the clinic's own server.

"Can't a simple heat warning do this?" E4 answers it: we compare directly with a heat-index rule. And a heat warning treats everyone the same; two patients in the same heatwave get very different risk in our twin.

"Has this been done before?" Personal heat-strain twins exist for healthy workers and soldiers (NTU Singapore, the ThermRisk project). We didn't find one for chronically ill patients that uses the EHR, and specifically corrects for beta-blockers. That's our contribution.

Things to verify on Day 1

These were written from knowledge, not checked against the original papers; confirm each before coding against it.

- [ ] **PSI formula** (Moran, Shitzer & Pandolf, 1998, *Am J Physiol*): the 39.5°C and 180 bpm constants and the 5 + 5 weighting
- [ ] **PSI bands** (which scores count as "high"), to justify the threshold of 7
- [ ] **Buller et al. 2013 core-temperature Kalman filter** (*Physiological Measurement*): the quadratic HR-to-core-temp coefficients, process and observation noise values, and starting core temperature
- [ ] **Drug effects in heat:** a review or guideline page on diuretics, SGLT2 inhibitors and beta-blockers in hot weather, to cite in the README

- [ ] **Synthea:** that its output includes the conditions, medications and lab values we need; else switch to the fallback generator
- [ ] **Open-Meteo historical API:** the variables and Indian cities/dates you'll use
- [ ] **Unstop page:** recheck the submission rules and deadline the night before

Sources

- Digital Twin Challenge 2026: Unstop listing
- Wearable Sensor Technology to Predict Core Body Temperature: A Systematic Review
- NTU: AI-based human digital twins for worker heat safety
- ThermRisk: personal digital twin for thermal risk (HVL)
- US Patent 11564579: adaptive Physiological Strain Index